Solution: Topological Sort

`Topological Sort` is used to find a linear ordering of elements that have dependencies on each other. For example, if event ‘B’ is dependent on event ‘A’, ‘A’ comes before ‘B’ in topological ordering.

This pattern defines an easy way to understand the technique for performing topological sorting of a set of elements and then solves a few problems using it.

Let’s see this pattern in action.

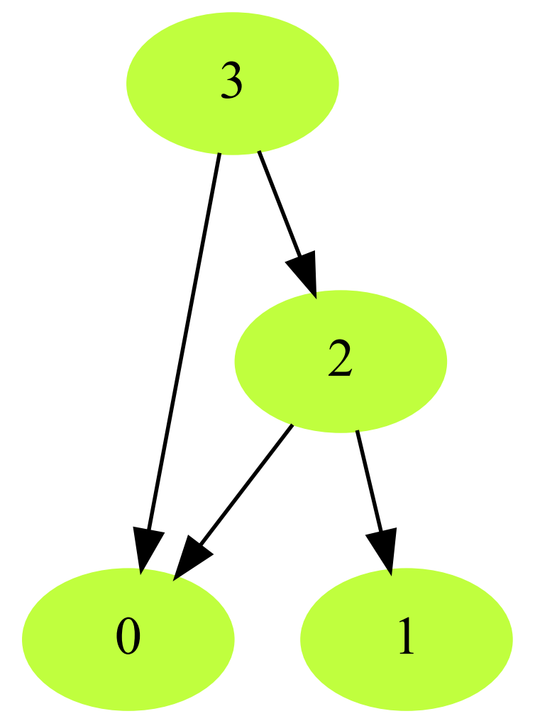
Problem Statement

Topological Sort of a directed graph (a graph with unidirectional edges) is a linear ordering of its vertices such that for every directed edge (U, V) from vertex U to vertex V, U comes before V in the ordering.

Given a directed graph, find the topological ordering of its vertices.

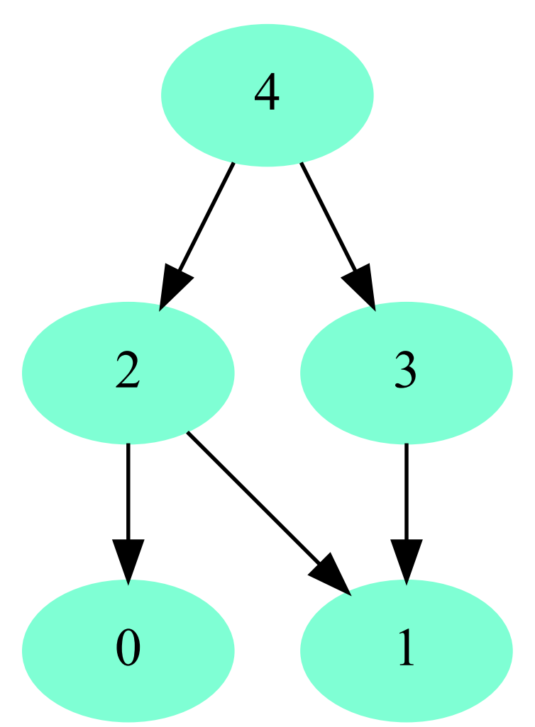
Example 1:

```
Input: Vertices=4, Edges=[3, 2], [3, 0], [2, 0], [2, 1]
Output: Following are the two valid topological sorts for the given
graph:
1) 3, 2, 0, 1
2) 3, 2, 1, 0
```



Example 2:

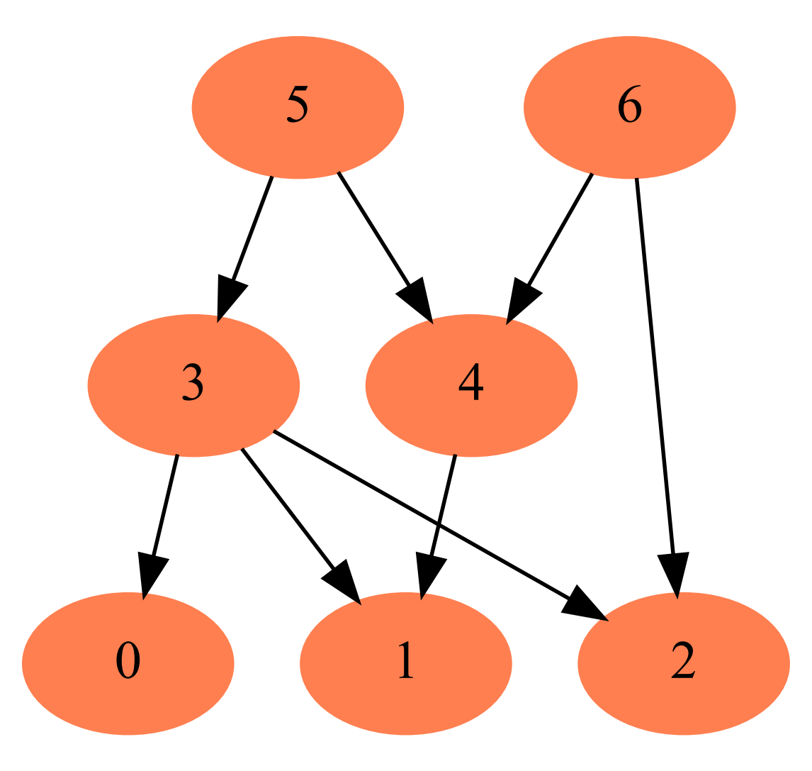
```
Input: Vertices=5, Edges=[4, 2], [4, 3], [2, 0], [2, 1], [3, 1]
Output: Following are all valid topological sorts for the given graph:
1) 4, 2, 3, 0, 1
2) 4, 3, 2, 0, 1
3) 4, 3, 2, 1, 0
4) 4, 2, 3, 1, 0
5) 4, 2, 0, 3, 1
```



Example 3:

```
Input: Vertices=7, Edges=[6, 4], [6, 2], [5, 3], [5, 4], [3, 0], [3, 1],
[3, 2], [4, 1]
Output: Following are all valid topological sorts for the given graph:
1) 5, 6, 3, 4, 0, 1, 2
2) 6, 5, 3, 4, 0, 1, 2
3) 5, 6, 4, 3, 0, 2, 1
4) 6, 5, 4, 3, 0, 1, 2
5) 5, 6, 3, 4, 0, 2, 1
6) 5, 6, 3, 4, 1, 2, 0

There are other valid topological ordering of the graph too.
```



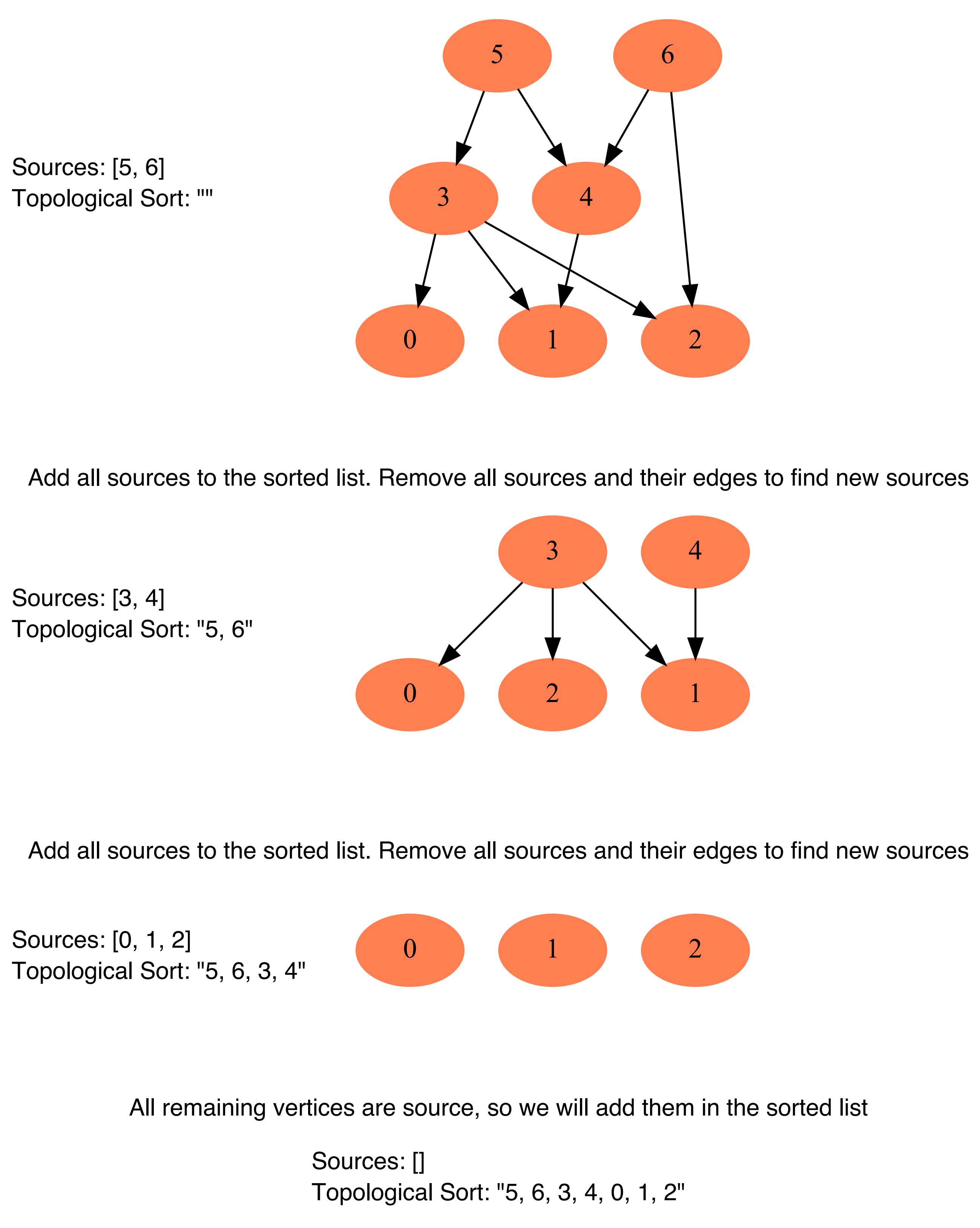
Solution

The basic idea behind the topological sort is to provide a partial ordering among the vertices of the graph such that if there is an edge from `U` to `V` then U≤V i.e., `U` comes before `V` in the ordering. Here are a few fundamental concepts related to topological sort:

1. **Source**: Any node that has no incoming edge and has only outgoing edges is called a source.
2. **Sink**: Any node that has only incoming edges and no outgoing edge is called a sink.
3. So, we can say that a topological ordering starts with one of the sources and ends at one of the sinks.
4. A topological ordering is possible only when the graph has no directed cycles, i.e. if the graph is a **Directed Acyclic Graph** (**DAG**). If the graph has a cycle, some vertices will have cyclic dependencies which makes it impossible to find a linear ordering among vertices.

To find the topological sort of a graph we can traverse the graph in a **Breadth First Search** (**BFS**) way. We will start with all the sources, and in a stepwise fashion, save all sources to a sorted list. We will then remove all sources and their edges from the graph. After the removal of the edges, we will have new sources, so we will repeat the above process until all vertices are visited.

Here is the visual representation of this algorithm for Example-3:


Sources: [5, 6]
Topological Sort: ""
5
6
3
4
0
1
2
Add all sources to the sorted list. Remove all sources and their edges to find new sources
Sources: [3, 4]
Topological Sort: "5, 6"
3
4
0
2
1
Add all sources to the sorted list. Remove all sources and their edges to find new sources
Sources: [0, 1, 2]
Topological Sort: "5, 6, 3, 4"
0
1
2
All remaining vertices are source, so we will add them in the sorted list
Sources: []
Topological Sort: "5, 6, 3, 4, 0, 1, 2"

This is how we can implement this algorithm:

a. Initialization

1. We will store the graph in **Adjacency Lists**, which means each parent vertex will have a list containing all of its children. We will do this using a **HashMap** where the ‘key’ will be the parent vertex number and the value will be a **List** containing children vertices.
2. To find the sources, we will keep a **HashMap** to count the in-degrees i.e., count of incoming edges of each vertex. Any vertex with ‘0’ in-degree will be a source.

b. Build the graph and find in-degrees of all vertices

1. We will build the graph from the input and populate the in-degrees **HashMap**.

c. Find all sources

1. All vertices with ‘0’ in-degrees will be our sources and we will store them in a Queue.

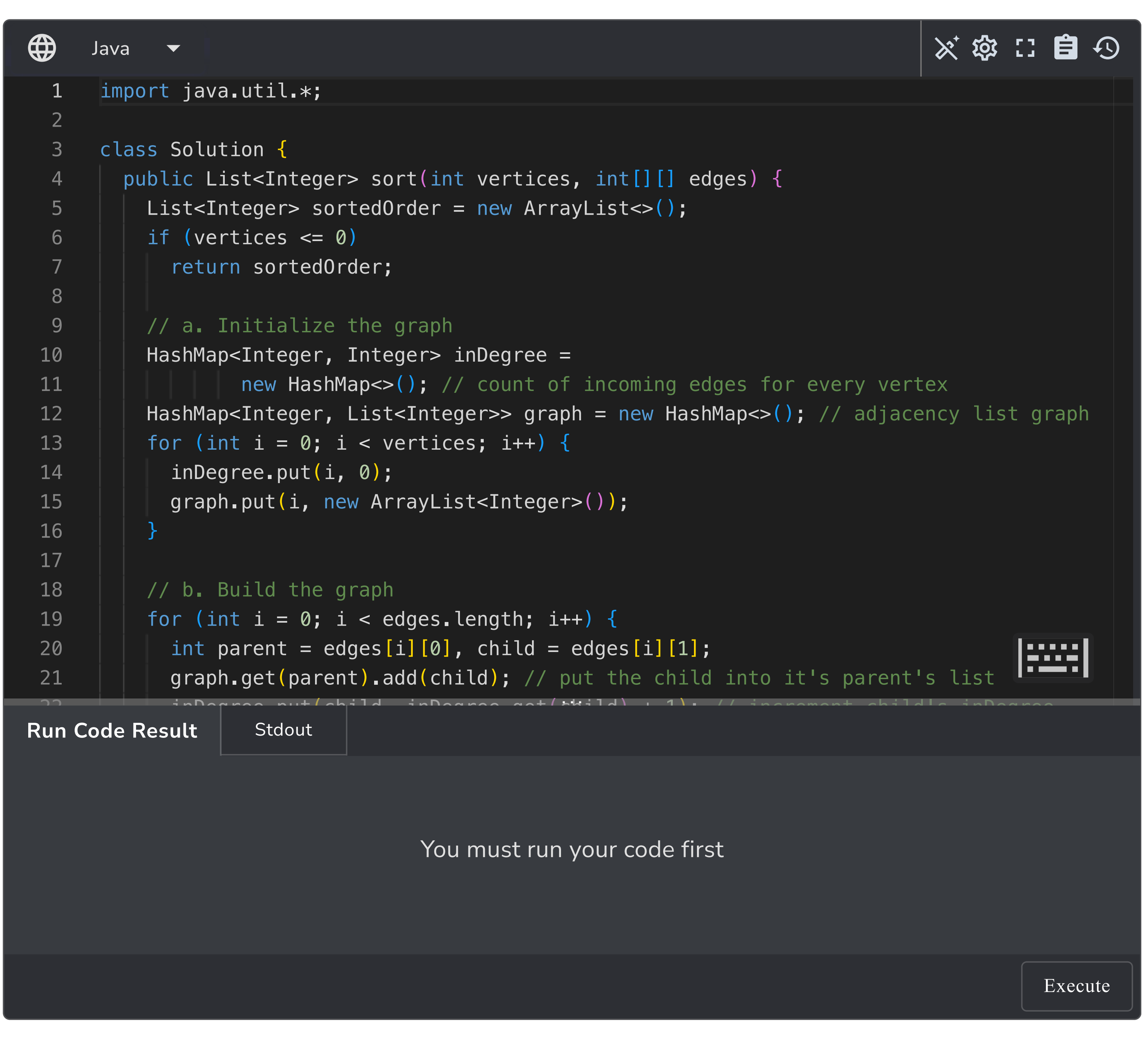
d. Sort

1. For each source, do the following things: Add it to the sorted list. Get all of its children from the graph. Decrement the in-degree of each child by 1. If a child’s in-degree becomes ‘0’, add it to the sources Queue.

2. Repeat step 1, until the source Queue is empty.

Code

Here is what our algorithm will look like:



```java
import java.util.*;

class Solution {
  public List<Integer> sort(int vertices, int[][] edges) {
    List<Integer> sortedOrder = new ArrayList<>();
    if (vertices <= 0)
      return sortedOrder;

    // a. Initialize the graph
    HashMap<Integer, Integer> inDegree =
            new HashMap<>(); // count of incoming edges for every vertex
    HashMap<Integer, List<Integer>> graph = new HashMap<>(); // adjacency list graph
    for (int i = 0; i < vertices; i++) {
      inDegree.put(i, 0);
      graph.put(i, new ArrayList<Integer>());
    }

    // b. Build the graph
    for (int i = 0; i < edges.length; i++) {
      int parent = edges[i][0], child = edges[i][1];
      graph.get(parent).add(child); // put the child into it's parent's list
```

You must run your code first

Time Complexity

In step ‘d’, each vertex will become a source only once and each edge will be accessed and removed once. Therefore, the time complexity of the above algorithm will be O(V+E), where ‘V’ is the total number of vertices and ‘E’ is the total number of edges in the graph.

Space Complexity

The space complexity will be `O(V+E)`, since we are storing all of the edges for each vertex in an adjacency list.

Similar Problems

Problem 1: Find if a given Directed Graph has a cycle in it or not.

Solution: If we can’t determine the topological ordering of all the vertices of a directed graph, the graph has a cycle in it. This was also referred to in the above code:

```
    if (sortedOrder.size() ≠ vertices) // topological sort is not possible as the graph
has a cycle
      return new ArrayList<>();
```

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